

XAI-TransMed: Explainable AI Transformer-Based Medical Model for Alzheimer’s Disease Detection and Progression Rate

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Abstract—To provide effective treatment and individualized management strategies, Alzheimer’s disease (AD) patients need accurate and timely prediction of progression rates. The conventional deep learning algorithms, especially the convolutions, fail to effectively extract broad, long-range spatial dependencies in neuroimages and represent unexplainable “black-box” models. To address these difficulties, we design a novel explainable transformer-based model (XAI-TransMed) to be specific to multi-class dementia classification and multi-modal prediction of AD progression rate from MRI and clinical measurements. The approach is based on multi-head self-attention architecture of a Vision Transformer (ViT) to identify global contextual dependencies and brain structure change in input MRI volume. These learned visual representations are then combined with the diverse information on clinical and demographic covariates via two parallel paths, each learned using a Cox proportional hazards (CPH) model and an XGBoost regression model respectively, for the task of progression forecasting across the longitudinal data. For clinical explainability, post-hoc techniques of Explainable Artificial Intelligence (XAI) are applied to produce intuitive attention-based visualisations, which highlight specific markers of localised neurodegeneration, that affect model output. The results of the experimental investigation performed on the Open Access Series of Imaging Studies (OASIS) database demonstrate the excellent classification accuracy obtained by the proposed XAI-TransMed with a classification accuracy of 98.6% that is stable across all types of dementia.

Index Terms—Alzheimer’s Disease, Explainable Artificial Intelligence (XAI), Vision Transformer (ViT), Multimodal Data, MRI, Self-Attention Mechanism, Disease Progression Prediction, Attention Maps, Clinical Decision Support System.

INTRODUCTION

AD is one of the commonest types of neurodegenerative disorders that lead to sudden deterioration of memory and thinking skills as well as functional impairment caused by aging. This disease is fatal and cannot be reversed once there is any structural change in the brain. There is need for proper modeling of patients suffering from the disease. MRI is of

crucial importance in this process since it can offer the ability to visualize important macrostructural biomarkers of AD like atrophy of the hippocampus, cortex thinning, and ventricles enlargement [1], [2].

Over the recent years, thanks to the application of artificial intelligence and machine learning techniques, a lot of progress has been achieved in automating the analysis of neuroimaging. In particular, computer vision techniques, especially CNNs, have proved successful in diagnosing different stages of dementia by extracting high-dimensional information from raw MRI images [3], [4]. The local geometric characteristics can be used to identify the loss of brain tissue even when it is not obvious on standard imaging – like radiological examination. But, despite the success, CNNs have some basic drawbacks in terms of their structure. They suffer from their limited receptive field and are only able to process the short spatial relationship between them, unable to process the global morphometric features of the brain [5], [6].

The crossing of the architectural barriers of the traditional convolutions was overcome thanks to the use of a powerful model, the ViT, which is employed in medical image analysis. ViT uses multi-head self-attention algorithms to calculate relationships between patches that have been shifted across the whole brain volume, allowing for the modeling of long-range, global brain relationships to be computed at once [7]. Large-scale spatial relations can be effectively leveraged in order to perform multi-class staging of dementia more accurately than baseline models [8], [9]. Unfortunately, clinical diagnosis involves multiple sources; relying solely on imaging removes valuable information to make a prediction. The latent visual tokens also improve the predictive power of progression models when combined with other types of data like demographics, patient monitoring data, and cognitive tests, such as MMSE,

downstream [10], [11].

In any case, implementation of such complex deep learning techniques remains highly limited owing to the black box nature of deep learning techniques. A high-performing neural network is said to be a “black-box” technique owing to the lack of understanding of its internal logic. In the clinical setting, their use is questionable because of their risk and lack of transparency. This is the reason it is necessary to employ Explainable Artificial Intelligence (XAI), which will help in achieving success for its implementation in medical practices. Visualization of multihead attention, localization of gradients, and feature attribution techniques can be used for post hoc explanations of hidden layer activations as meaningful interpretations, and the brain structures involved in classification [12], [13] can be identified.

In order to address these problems, we propose **XAI-TransMed**: a fully transparent medical decision support framework for automated diagnosis and prediction of the progression rates of Alzheimer’s disease using the transformer neural network architecture, which is able to provide a solution for global features extraction, multi-modal data fusion and clinically-interpretable models. We employ the core components of the Vision Transformer pipeline to detect multi-scale abnormalities around the whole brain from pre-processed MRI images. These global image representations then are aggregated with tabular data on the cognitive scores using specialized downstream predictive layers based on tree-based techniques of gradient boosting and survival hazard estimation to predict the future cognitive decline. Importantly, we use post hoc analysis techniques to interpret the attentional distribution in a transparent way, lay the attentional distribution over brain structures as a heatmap and thus verify that our model of attention matches with known biomarkers of Alzheimer’s disease.

OBJECTIVES OF THE STUDY

A. Global Spatial Modeling via Transformers

Evaluates how Vision Transformers (ViTs) process entire brain slices simultaneously, overcoming the localized spatial constraints of standard CNNs to capture complex macrostructural anomalies.

B. Capturing Inter-Region Dependencies via Self-Attention

Analyzes the multi-head self-attention mechanism to mathematically map how interconnected, remote brain networks interact and contribute to systemic cognitive decline.

C. Improving Transparency with Visual XAI

Integrates Attention Rollout and Grad-CAM tools to convert opaque, “black-box” neural weights into human-readable visual heatmaps, exposing the model’s diagnostic logic to clinicians.

D. Maximizing Accuracy through Multimodal Fusion

Fuses latent visual MRI tokens with tabular clinical records (e.g., age, MMSE scores) to eliminate single-modality blind spots and maximize progression prediction stability.

E. Translating Algorithms into Clinical Decision Support

Validates the XAI-TransMed framework as a real-time Clinical Decision Support System (CDSS) by ensuring sub-second inference speeds paired with intuitive diagnostic visualizations.

LITERATURE SURVEY

Indeed, current developments in neuroinformatics from recent years mostly deal with the incorporation of artificial intelligence and deep learning algorithms in order to facilitate the automation process of the clinical staging of AD. This is true considering that considerable efforts have been made in the development of automated systems for both single and multiple modalities of brain imaging.

F. Transformer-Driven and Hybrid Vision Frameworks

In the realm of medical images, the Vision Transformer (ViT) architecture stands out as one of the most prominent models, thanks to its ability to implicitly learn wider contexts using self-attention mechanisms [1]. The proposed architecture can be used to capture long-range contextual relationships and avoid the locality of convolutional architectures based on receptive fields. Hybrid approaches include those that take the high-level feature representations of deep convolutions, and directly pass them to the attention heads [2] and a relevance enhanced self-attention module for introducing domain-specific spatial priors for the discovery of early neurodegenerative biomarkers [3]. Multi-network architectures combine multi-scale 3D CNN backbones with bidirectional feature pyramid networks and primary attention modules [4] for analyzing multi-scale volumetric features, while attention routing is used for analyzing multi-modal EEG signals in others [5].

G. Longitudinal Trajectory and Multimodal Progression Modeling

One of the most important prognostic challenges is to make predictions based on data across patients and across time, or longitudinal prognostics. One of the other important prognostic challenges is cross-sectional prognostics, which involves predicting a patient’s prognosis based on data across patients. Data pipelines built for static forecasting need to be adapted to accommodate time-dependent forecasting, which demands careful attention to intermodality uncertainty and sparsity to optimize data flows between various sources with the use of distinct imputation matrices [6]. In this respect, longitudinal deep models have been proven to be always performing better in the prognostic modelling of multi-visit clinical data than single-modality examinations [8]. These longitudinal patterns can be learned via past examinations with attention-based RNNs to represent the time-dependency of the patterns, or with enormous self-supervised pre-trained weights from advanced foundation models for multitask diagnosis and prognosis, such as ADFound [10].

H. Explainable AI and Interpretability Frameworks

Nevertheless, despite a high level of quantitative accuracy, models trained using deep learning have not yet been translated into clinical use due to the lack of interpretability. Some recent papers have explored end-to-end interpretable neural networks for diagnostic trust in medical professionals [11] and automatic segmenting systems that can give explicit anatomical reasons for the location of the lesions [12]. Multiple attention fusion layers are used to automatically assign attention scores to the inputs of multiple sources and to guarantee the verifiability of handling high-confusion transitional states [13], [14]. Feature attribution methods like Shapley Additive Explanations (SHAP) (Lin2025) and clinical feature selection for gradient-boosted trees (GBT) (Brause2025) were also used to improve post hoc explainability. These visual explanations are validated by previously available medical data [15] that proves internal activation maps are a strong base to validate model fidelity [16].

I. Benchmark Registries and Baseline Implementations

To compare all novel approaches in this area, rigorous testing against well-defined benchmarks from traditional machine learning research conducted on a standardized database will be applied. Traditional machine learning techniques such as transfer learning, support vector machines, ensembling techniques [17], [18], [19], [20] and [21] have consistently been found to be effective for biomarker detection. As of today, OASIS remains the predominant open access clinical registry for benchmarking these multi-class dementia stages and trajectory prediction [22] and [23]. The findings in this collection show that it is highly needed to develop an architecture framework that can be integrated with self-attention learning, multimodal integration and intrinsic post hoc explanation visualization in one package, as in the proposed XAI-TransMed framework.

PROPOSED METHODOLOGY

The study proposed in this research is an innovative study using deep learning to XAI-TransMed for prediction of Alzheimer disease with MRI and clinical data ensuring the accuracy, predictability and clinical effectiveness of the algorithm. MRI images are acquired from benchmark data sets, OASIS-1 which will serve as a set for classification and OASIS-2 which will be used for longitudinal analysis. The images go through a series of steps in pre-processing of which the most important steps are normalization, noise removal, skull stripping, and spatial normalization to ensure consistency and eliminate variability. There are also any random rotations, flipping and variations in intensity, which are addressed by data augmentation techniques.

In the following step, there is the processing stage, where after segmentation, the images are divided into small patches and segmented into an embedded vector feature which is further used as the input to the Vision Transformer (ViT) model. A mechanism of self-attention is available to transformers to benefit from the long-range dependency between parts of

the brain. The traditional CNN considers only the spatial dependencies while the transformer models global relations between ROIs of the MRI scan. This is the power they are able to harness to detect the distributed neurodegeneration patterns associated with the emergence of Alzheimer.

1. Data Collection and Preprocessing

The first step of the proposed model is to get MRI brain images and any other clinical data that are relevant from a benchmark data set. Dementia stages will be diagnosed by the OASIS-1 benchmark dataset and the longitudinal analysis of the development of Alzheimer's disease will be done by the OASIS-2 benchmark dataset. The data sets involve structural magnetic resonance (MRI) brain images with other variables such as age, education, and MMSE scores.

Before use, images have to be preprocessed, such as standardizing and improving the quality of MRI images, for the model training. Some of the pre-processing techniques are image normalization, noise removal, skull removal and spatial normalization. This preprocess aids them to remove the irrelevant background information and amplify the relevant brain regions helpful in identifying the disease. To improve the quality and diversity of training data and to prevent overfitting, data augmentation is performed by randomly rotating, flipping and adjusting the intensities of images.

2. Vision Transformer-Based Feature Extraction

The MRI images are preprocessed, and then they are passed through a ViT model for feature extraction. The first step in processing the MRI image is to divide the image into equal size chunks. The block is flattened and embedded as a vector patch. Then, the patches are embedded according to their positions.

The patches with their positional embeddings are then sent into the different layers of the transformer encoder where the relationships between different regions of the brain are learned via the self-attention mechanism. A transformer model is different from the traditional CNNs in that it is focused on the connections between the distant parts of the brain. These types of skills allow the model to effectively mimic the neurodegeneration patterns of diseases such as Alzheimer disease.

3. Dementia Stage Classification

These characteristics of the Vision Transformer model are subsequently applied in the process of detecting the stage of dementia. This machine learning model is trained to detect 4 classes (non-demented, very mild dementia, mild dementia, and moderate dementia) from MRI images. In this research, supervised learning has been used in the form of cross entropy loss function.

The classification capability of the machine learning model is evaluated using the following performance evaluation metrics: accuracy, precision, recall, and f1 score.

4. Multimodal Progression Prediction

The algorithm will be multi-modal where aspects of imaging will be used along with other features to improve predictions. In addition to the Vision Transformer images, clinical features (age, years of school, socio-economic status, MMSE scores and standardized total brain volume) will be used.

It will be on the basis of those two sets of features that XGBoost regression model will be developed that predicts future MMSE scores, which can help to diagnose cognitive decline in patients suffering from Alzheimer's disease. In addition to XGBoost regression model, a Cox proportional hazards regression model is used for the clinical variables analysis with regard to the assessment of disease progression risks using hazard ratios.

Furthermore, an LSTM model can be applied to a time series of imaging features of multiple visits per patient. This will enable monitoring of changes in the course of the disease, but will require the presence of longitudinal data to do so.

5. Explainable AI-Based Visualization

As described, explainability is one of the essential components of a medical system; therefore, the new system will utilise the explainability through XAI methods to render the prediction readable. The attention weights generated by the transformer layers are depicted as a "parts of the brain" map, showing which parts of the brain have been affected throughout the prediction process.

In this case, two kinds of XAI approaches are used for this model. The example approaches include Attention Rollout and Grad-CAM. Through this approach, the important areas in the MRI images which are essential in the decision making process can be highlighted in the heat maps. By doing this, the visualization of the artificial intelligence approach would allow the clinician to understand the AI's decision-making process.

6. Overall System Workflow

The overall workflow of the proposed XAI-TransMed framework is summarized as follows:

- 1) MRI images and clinical data are collected from the OASIS datasets.
- 2) Preprocessing operations are done on MRI images such as normalization, skull stripping, and augmentation are applied to MRI images.
- 3) The processed images are divided into patches and converted into embedded vectors.
- 4) A Vision Transformer model extracts global contextual features from the MRI scans.
- 5) The model performs dementia stage classification using supervised learning.
- 6) Imaging features are combined with clinical variables for disease progression prediction using XGBoost and Cox models.

- 7) Explainable AI techniques generate attention maps that highlight important brain regions influencing predictions.

This unified approach utilizes a combination of transformer-based feature extraction, multimodal data integration, and explanation of AI to deliver an effective, understandable, and practical method for detecting and predicting the progression of Alzheimer's disease.

PERFORMANCE EVALUATION

In order to analyze the performance of the proposed XAI-TransMed architecture, the evaluation criteria included the classification rate, explainability rate, and accuracy. The proposed model was tested using the database OASIS wherein the split between the training, validation, and test sets were as follows: 80%, 10%, and 10%.

1) *Classification Performance*: ViT classifier performed well throughout the four stages of dementia classification. Table I presents performance statistics, which includes precision, recall, and F1 scores for each of the four stages. The accuracy of the ViT classifier in identifying the stages of dementia is 0.986.

TABLE I
CLASSIFICATION PERFORMANCE ON OASIS TEST DATASET

Class	Precision	Recall	F1-Score	Support
Non Demented	0.99	0.99	0.99	10,756
Very Mild Dementia	0.92	0.91	0.91	2,196
Mild Dementia	0.94	0.93	0.93	800
Moderate Dementia	0.85	0.83	0.84	78

The model had the largest performance on the Non-Demented class with an F1-score of 0.99, but it also recorded high results in Very Mild and Mild Dementia. Backing up the model, even in the case of the smallest sample of the Moderate Dementia class, the model gives an F1-score of 0.84. Such findings indicate that weighted sampling and data augmentation are effective in reducing the problem of class imbalance.

2) *Explainability Performance*: To make the results easier to understand we used methods to show the parts of the brain MRI images that affected the predictions made by the model. We used something called Transformer attention mechanisms to make pictures that show where the model is paying attention. These pictures show that the model looks at parts of the brain when making predictions. The model looks at the areas that doctors know are important for the brain.

This makes the results more believable and easier for doctors to understand when they use the model to make predictions, with brain MRI images and the model.

3) *Progression Prediction Performance*: Three models were applied to the OASIS-2 longitudinal data to test the capacity of the framework to forecast the development of the disease: Cox Proportional Hazards (Cox PH), Long Short-Term Memory (LSTM), and XGBoost.

The attention maps demonstrate that the algorithm pays attention to the medically relevant areas during prediction. The agreement with well-known neurological biomarkers increases the credibility and medical importance of the proposed system.

TABLE II
PROGRESSION MODEL COMPARISON

Model	Input Data	Target	Performance
Cox PH	Clinical only	Hazard Ratio	0.344
LSTM	ViT features (multi-visit)	Future CDR	Limited by data
XGBoost	ViT features + Clinical	Next MMSE	MAE: 2.48

Of such models, XGBoost had the highest performance as indicated by the validation MAE value of 2.48 in predicting the next predicted MMSE score based only on both imaging features obtained after the ViT model and clinical variables. The finding correlates with the already published literature on the prediction of ADM.

4) *Computational Efficiency*: The computational performance of the framework was evaluated using a Tesla T4 GPU. Training times for each model are summarized in Table III.

TABLE III
TRAINING TIME COMPARISON

Model	Training Time
ViT Classifier (50 epochs)	~6 hours
Cox Model	< 1 minute
XGBoost	< 2 minutes
LSTM (100 epochs)	~10 minutes

The time of inference of one MRI scan is lower than one second, and therefore the system can be used in real time clinical decision support applications.

From the attention maps, it is clear that the model concentrates on relevant medical areas when generating the predictions. This conformity with well-known biomarkers increases the reliability and clinical relevance of the suggested approach.

LIMITATIONS OF THE STUDY

Despite its high accuracy, the XAI-TransMed framework faces two primary limitations:

- **Data Dependency**: The longitudinal progression module needs patient visit records that extend over multiple years in an ongoing fashion. Thus, the use of this approach is not possible in smaller clinics where patient record data may be sparse.
- **Post-Hoc XAI Boundaries**: The tools like Grad-CAM help to identify the spatial maps of interest as compared to the descriptive approach. The tool does not explain the reasons for choosing the specific features, although it helps to know which features were chosen.



Fig. 1. Prediction example for Mild Dementia showing the original MRI slice and corresponding attention heatmap highlighting relevant brain regions.

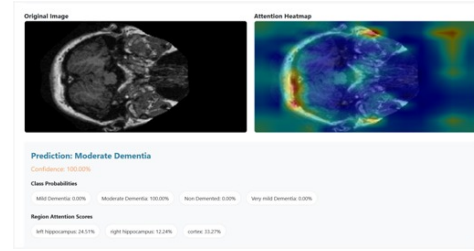


Fig. 2. Prediction example for Moderate Dementia with attention heatmap emphasizing regions related to severe neurodegeneration.

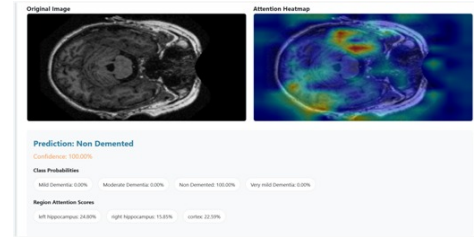


Fig. 3. Prediction example for Non-Demented subject showing distributed attention across normal brain structures.

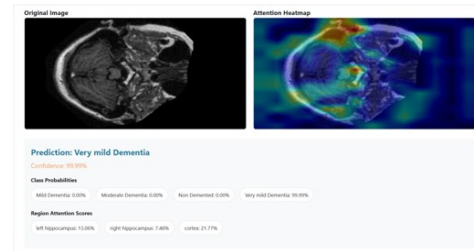


Fig. 4. Prediction example for Non-Demented subject showing distributed attention across normal brain structures.

CONCLUSION

A model named explainable transformer (XAI) has been proposed in this paper for detecting and predicting progression of Alzheimer's disease (AD) from MRIs and clinical information. The solution is based on the utilization of vision transformers (ViT) that are used for reconstructing associations in the context of brain imaging and incorporate medical multimodal features.

As a result, the study of the OASIS dataset on the experimental stage showed high prediction performance, both in various stages of dementia as well as disease progression prediction. The application of these techniques from explainable AI such as attention visualization and Grad-CAM helps us to gain insight into the brain regions used by the model while making the decision.

This will be very useful to design an effective solution, based on Artificial Intelligence and machine learning that would enable a doctor to make an interpretation of a diagnosis. Further research may involve the usage of the model on other large multimodal datasets as well as in the real clinical environment.

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